

Advanced Dynare

Chapter XIV

Optimal Policy and Welfare Assessment

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Outline

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2. Optimal policy under commitment
 - Theory
 - Practice
3. Discretionary policy
 - Theory
 - Practice
4. Optimal simple rules
 - Theory
 - Practice

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Motivation

- Policy-makers are often interested in implementing some form of optimal policy
- Typical problem structure involves:
 1. optimizing some welfare or loss criterion,
 2. taking into account private sector reactions to policy changes
 3. while being constrained by technologies, institutions, etc.
- Welfare analysis allows
 - model-consistent comparison of policies and institutions
 - preventing socially inefficient equilibria

Rules vs. discretion

- In dynamic models, private sector expectations about government behavior matter for private decision-making (e.g. Lucas 1976)
- However: government may have incentive to renege on earlier promises made to private sector (time-consistency problem of Kydland and Prescott (1977))
- If private sector anticipates renegeing, it will adjust its behavior accordingly
→ policy-tradeoffs like sacrifice ratio can become more expensive
- Solution may be credibly committing to future policy actions

Two important conceptual distinctions

- First important conceptual distinction between
 1. **discretionary policy**
 - policy-maker re-optimizes period-by-period
 2. optimal policy under **commitment**
 - announce a policy plan and stick to it no matter what
- Policy under commitment in turn can be classified in two kinds of policies
 1. fully state-contingent, i.e. **Ramsey** optimal policy
 - first-best policy that can optimally react to all state variables including shocks
 2. **optimal simple rules (OSR)**
 - policy rule restricted to a specific functional form (e.g. Taylor rule or fiscal feedback)
- OSR is easier to communicate and implement, but may only be associated with a small welfare loss compared to Ramsey policy (e.g. Schmitt-Grohé and Uribe 2007)

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General case

- The generic Ramsey problem can be written as

$$\max_{\{y_t\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(y_t), \quad (1)$$

where $U()$ is the planner objective and β the planner discount factor

- Optimization is conducted subject to the private sector equilibrium conditions:

$$\mathbb{E}_t f(y_{t+1}, y_t, y_{t-1}, \varepsilon_t) = 0, \quad \forall t \geq 0, \quad (2)$$

taking y_{-1} as given and imposing suitable transversality conditions, where

- $y_t \in \mathbb{R}^n$: endogenous variables
 - $\varepsilon_t \in \mathbb{R}^p$: stochastic innovations
 - $f : \mathbb{R}^{3n+p} \rightarrow \mathbb{R}^m$
- There are $n - m$ free policy instruments

The Ramsey planner's Lagrangian

- Denoting with $\mu_t \in \mathbb{R}^m$ the vector of *discounted* Lagrange multipliers, the Lagrangian of the Ramsey problem is given by

$$\mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t (U(y_t) - \mu_t' f(y_{t+1}, y_t, y_{t-1}, \varepsilon_t)) \quad (3)$$

- Dynare will automatically construct this Lagrangian internally and compute the necessary first order conditions if a `ramsey_model` statement is present
- Resulting first-order conditions can be inspected using Dynare's $\text{\LaTeX}$ export functionality (`write_latex_dynamic_model`)

First order necessary conditions

- The first order necessary conditions (FOC) for an optimum are given by

$$0 = \frac{\partial \mathcal{L}}{\partial \mu_t} = \mathbb{E}_0 f(y_{t+1}, y_t, y_{t-1}, \varepsilon_t), \quad t \geq 0 \quad (4)$$

$$0 = \frac{\partial \mathcal{L}}{\partial y_0} = \mathbb{E}_0 \left\{ U'(y_0) - \mu'_0 \frac{\partial f}{\partial y_t} \Big|_{(y_1, y_0, y_{-1}, \epsilon_0)} - \beta \mu'_1 \frac{\partial f}{\partial y_{t-1}} \Big|_{(y_2, y_1, y_0, \epsilon_1)} \right\} \quad (5)$$

$$0 = \frac{\partial \mathcal{L}}{\partial y_t} = \mathbb{E}_0 \left\{ U'(y_t) - \beta^{-1} \mu'_{t-1} \frac{\partial f}{\partial y_{t+1}} \Big|_{(y_t, y_{t-1}, y_{t-2}, \epsilon_{t-1})} - \mu'_t \frac{\partial f}{\partial y_t} \Big|_{(y_{t+1}, y_t, y_{t-1}, \epsilon_t)} - \beta \mu'_{t+1} \frac{\partial f}{\partial y_{t-1}} \Big|_{(y_{t+2}, y_{t+1}, y_t, \epsilon_{t+1})} \right\}, \quad t > 0 \quad (6)$$

- First period is special, rendering the problem non-time invariant (and time-inconsistent)!
- However: (6) reduces to (5) with the initial Lagrange multiplier set to $\mu_{-1} = 0$

Inspecting the structure

- With $\mu_{-1} = 0$, we have a time-invariant non-linear system of rational expectation equations for all t :

$$0 = \frac{\partial \mathcal{L}}{\partial \mu_t} = \mathbb{E}_0 f(y_{t+1}, y_t, y_{t-1}, \varepsilon_t), \quad (7)$$

$$0 = \frac{\partial \mathcal{L}}{\partial y_t} = \mathbb{E}_0 \left\{ U'(y_t) - \beta^{-1} \mu'_{t-1} \frac{\partial f}{\partial y_{t+1}} \Big|_{(y_t, y_{t-1}, y_{t-2}, \epsilon_{t-1})} - \mu'_t \frac{\partial f}{\partial y_t} \Big|_{(y_{t+1}, y_t, y_{t-1}, \epsilon_t)} - \beta \mu'_{t+1} \frac{\partial f}{\partial y_{t-1}} \Big|_{(y_{t+2}, y_{t+1}, y_t, \epsilon_{t+1})} \right\}, \quad (8)$$

given initial and transversality conditions

- System is amenable to both perfect foresight solver and k-order perturbation methods

Steady state

- Perturbation of the non-linear system can be accurate because, in the absence of shocks, the system may converge to a well-defined steady state
- However: need to find a steady state for both μ and y first
- The steady-state values $\bar{y}$ and $\bar{\mu}$ need to satisfy

$$f(\bar{y}, \bar{y}, \bar{y}, 0) = 0 \quad (9)$$

$$U'(\bar{y}) - \bar{\mu}' \left(\beta^{-1} \frac{\partial f}{\partial y_{t+1}} \Big|_{(\bar{y}, \bar{y}, \bar{y}, 0)} + \frac{\partial f}{\partial y_t} \Big|_{(\bar{y}, \bar{y}, \bar{y}, 0)} + \beta \frac{\partial f}{\partial y_{t-1}} \Big|_{(\bar{y}, \bar{y}, \bar{y}, 0)} \right) = 0 \quad (10)$$

- That is a potentially large non-linear system to solve!

Using a conditional steady state

- Problem can be simplified if a conditional steady state can be provided for y
- Take the planner instruments like the interest rate as given and compute the corresponding steady state of the private sector
→ implies a steady state candidate $\tilde{y}$
- Given $\tilde{y}$, we need to solve a linear equation to obtain $\tilde{\mu}$:

$$U'(\tilde{y}) - \tilde{\mu}' \underbrace{\left(\beta^{-1} \frac{\partial f}{\partial y_{t+1}} \Big|_{(\tilde{y}, \tilde{y}, \tilde{y}, 0)} + \frac{\partial f}{\partial y_t} \Big|_{(\tilde{y}, \tilde{y}, \tilde{y}, 0)} + \beta \frac{\partial f}{\partial y_{t-1}} \Big|_{(\tilde{y}, \tilde{y}, \tilde{y}, 0)} \right)}_M = U'(\tilde{y}) - \tilde{\mu}' M = 0 \quad (11)$$

- Iterate on value of instruments until convergence of $\tilde{y}$ and $\tilde{\mu}$ to $\bar{y}$ and $\bar{\mu}$
→ increases success probability of finding steady state to the problem
- However: steady state may not be unique
→ starting point for the instruments may matter

Welfare assessment

- We are often interested in explicitly evaluating the welfare function

$$\max_{\{y_t\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(y_t), \quad (12)$$

- Welfare at time t can be recursively written (making dependence on state variables x_{t-1} and shocks ε_t explicit):

$$W(x_{t-1}, \varepsilon_t) = U(x_{t-1}, \varepsilon_t) + \beta \mathbb{E}_t W(x_t, \varepsilon_{t+1}) \quad (13)$$

- Careful: welfare objective cannot be included in the model block as it will cause singularity
- `evaluate_planner_objective` allows evaluating this function
- Important: proper welfare assessment typically requires at least a second-order approximation of the model, unless there is an undistorted steady state (Woodford 2002)

Conditional vs. unconditional welfare

- `evaluate_planner_objective` reports two different welfare measures commonly of interest to policy-makers:

1. Conditional welfare at time $t = 0$:

$$W(x_{-1}, \varepsilon_0,) \quad (14)$$

- takes into account the dynamics starting from the initial state x_{-1} and the observed current shock ε_0
- allows accounting for transition behavior from old policy regime to new optimal policy regime
- Dynare reports value for two particular initial multipliers μ_{-1} states:
 - $\mu_{-1} = \bar{\mu}$ (steady state multiplier)
 - $\mu_{-1} = 0$ (no initial commitment by planner)

2. Unconditional welfare

$$\mathbb{E}[W(x_{t-1}, \varepsilon_t,)] = \frac{\mathbb{E}[U(x_{t-1}, \varepsilon_t,)]}{1 - \beta} \quad (15)$$

→ corresponds to the long-run expected value of welfare

In Dynare

- U is stated in the `planner_objective` command
- `ramsey_constraints` block contains constraints, e.g. ZLB
- `ramsey_model` command generates the FOC with options
 - `planner_discount`: central planner's discount factor
 - `instruments`: variables for the computation of the steady state under optimal policy
→ `steady_state_model` block or `_steadystate.m` file
- `initval`: initial instruments values
- `histval`: initial state variables values
- `shocks`: specifies ε_0
- call `stoch_simul` and `evaluate_planner_objective`

Dynare output

- `oo_.planner_objective_value` subfields:
 - `unconditional` contains $\mathbb{E}[W]$
 - `conditional` contains $W(x_{-1}, \varepsilon_0)$ in subfields
 - `steady_initial_multiplier` when $\mu_{-1} = \bar{\mu}$
 - `zero_initial_multiplier` when $\mu_{-1} = 0$
→ optimal policy surprise for private agents!

A New-Keynesian model

- Consider model based on Christiano et al. (2007)
→ shipped with Dynare (Ramsey_Example.mod)
- Representative household chooses consumption C_t and hours worked h_t to maximize discounted utility

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\log(C_t) - \frac{\chi}{2} h_t^2 \right] \quad (16)$$

- The budget constraint is given by

$$\frac{B_t}{P_t} = (1 + r_{t-1}) \frac{B_{t-1}}{P_t} - C_t + \frac{W_t}{P_t} h_t + \Theta_t, \quad (17)$$

where

- B_t : nominal private bonds (in zero net supply)
- r_t : nominal net interest rate
- W_t : nominal wage
- Θ_t : firm profits
- P_t : price level

Household FOCs

- The consumption FOC is given by the familiar Euler equation

$$\frac{1}{C_t} = \beta \mathbb{E}_t \left\{ \frac{1}{C_{t+1}} \frac{1 + r_t}{\Pi_{t+1}} \right\}, \quad (18)$$

where Π_t is the inflation rate

- The labor FOC can be written as

$$\chi h_t C_t = \frac{W_t}{P_t} \quad (19)$$

Firm setup

- There is a continuum of monopolistically competitive firms j with pricing power
- Production is linear in labor

$$C_{i,t} = e^{Z_t} h_{i,t}, \quad (20)$$

with TFP Z_t following

$$Z_t = \rho Z_{t-1} + u_t, u \stackrel{iid}{\sim} \mathcal{N}(0, \sigma_z^2) \quad (21)$$

- Firm j faces a downward-sloping demand curve for its variety

$$C_{j,t} = \left(\frac{P_{j,t}}{P_t} \right)^{-\theta}, \quad (22)$$

where θ is the Dixit-Stiglitz elasticity of substitution

- Price adjustment is subject to a Rotemberg (1982) quadratic cost

$$\frac{\phi}{2} \left(\frac{P_{j,t}}{P_{j,t-1}} - 1 \right)^2 C_{j,t} \quad (23)$$

Firm problem

- Firm j maximizes its after-tax profits

$$(1 + \tau) \frac{P_{j,t}}{P_t} C_{j,t} - MC_t C_{j,t} - \frac{\phi}{2} \left(\frac{P_{j,t}}{P_{j,t-1}} - 1 \right)^2 C_{j,t}, \quad (24)$$

where marginal cost is common to all firms:

$$MC_t = \frac{W_t}{P_t e^{Z_t}} \stackrel{(19)}{=} \frac{\chi h_t C_t}{e^{Z_t}} \quad (25)$$

- In a symmetric equilibrium, the pricing FOC is the well-known New Keynesian Phillips curve

$$\left[\tau - \frac{1}{\theta - 1} \right] (1 - \theta) + \theta \left(\frac{\chi h_t C_t}{e^{Z_t}} - 1 \right) = \phi (\Pi_t - 1) \Pi_t - \beta \mathbb{E}_t \phi (\Pi_{t+1} - 1) \Pi_{t+1} \quad (26)$$

Private sector equilibrium

- In equilibrium, the private sector FOCs given monetary policy can be represented as

$$\frac{1}{C_t} = \beta \mathbb{E}_t \left\{ \frac{1}{C_{t+1}} \frac{1+r_t}{\Pi_{t+1}} \right\} \quad (27)$$

$$\left[\tau - \frac{1}{\theta - 1} \right] (1 - \theta) + \theta \left(\frac{\chi h_t C_t}{e^{Z_t}} - 1 \right) = \phi(\Pi_t - 1)\Pi_t - \beta \mathbb{E}_t \phi(\Pi_{t+1} - 1)\Pi_{t+1} \quad (28)$$

$$e^{Z_t} h_t = C_t \left[1 + \frac{\phi}{2} (\Pi_t - 1)^2 \right] \quad (29)$$

$$Z_t = \rho Z_{t-1} + u_t \quad (30)$$

Ramsey problem

- The corresponding Ramsey problem is given by

$$\begin{aligned} \max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t & \left\{ \ln C_t - \frac{\chi}{2} h_t^2 - \mu_{1,t} \left(\frac{1}{C_t} - \beta \frac{1}{C_{t+1}} \frac{1+r_t}{\Pi_{t+1}} \right) \right. \\ & - \mu_{2,t} \left(\left[\tau - \frac{1}{\theta-1} \right] (1-\theta) + \theta \left(\frac{\chi h_t C_t}{e^{Z_t}} - 1 \right) \right. \\ & \quad \left. \left. - \phi(\Pi_t - 1) \Pi_t + \beta \phi(\Pi_{t+1} - 1) \Pi_{t-1} \right) \right. \\ & - \mu_{3,t} \left(e^{Z_t} h_t - C_t \left[1 + \frac{\phi}{2} (\Pi_t - 1)^2 \right] \right) \\ & \left. - \mu_{4,t} (Z_t - \rho Z_{t-1} - u_t) \right\} \end{aligned} \quad (31)$$

Dynare code

Ramsey_Example_commitment.mod

- We declare all 5 variables, including the interest rate set by the Ramsey planner:

```
var(log) C    $$C$ (long_name='Consumption')
      pi      $$\pi$ (long_name='Gross inflation')
      h       $$h$ (long_name='hours worked');

var Z         $$Z$ (long_name='TFP')
      r       $$r$ (long_name='Net nominal interest rate');

varexo epsilon ${\varepsilon}$ (long_name='TFP shock');
parameters beta  ${\beta}$ (long_name='discount factor')
      theta      ${\theta}$ (long_name='substitution elasticity')
      tau        ${\tau}$ (long_name='labor subsidy')
      chi        ${\chi}$ (long_name='labor disutility')
      phi        ${\phi}$ (long_name='price adjustment costs')
      rho        ${\rho}$ (long_name='TFP autocorrelation');
```

Dynare code (continued)

- The model only features 4 equations for 5 variables:

```
model;  
  [name='Euler equation']  
  1/(1+r)=beta*C/(C(+1)*pi(+1));  
  [name='Firm FOC']  
  (tau-1/(theta-1))*(1-theta)+theta*(chi*h*C/(exp(Z))-1)=  
    phi*(pi-1)*pi-beta*phi*(pi(+1)-1)*pi(+1);  
  [name='Resource constraint']  
  C*(1+phi/2*(pi-1)^2)=exp(Z)*h;  
  [name='TFP process']  
  Z=rho*Z(-1)+epsilon;  
end;
```

Dynare code (continued)

- We the steady state conditional on the interest rate r :

```
steady_state_model;
    Z=0;
    pi=(r+1)*beta;
    C=sqrt((1+1/theta*((1-beta)*(pi-1)*pi-(tau-1/(theta-1))*(1-theta)))/
        (chi*(1+phi/2*(pi-1)^2)));
    h=C*(1+phi/2*(pi-1)^2);
    y_nat=sqrt((theta-1)/theta*(1+tau)/chi);
    y_gap=log(C)-log(y_nat);
end;

shocks;
    var epsilon = 0.01^2;
end;
```

Dynare code (continued)

Finally, we set the initial value and the planner objective:

```
initval;  
    r=1/beta-1;  
end;  
planner_objective log(C)-chi/2*h^2;  
  
ramsey_model(instruments=(r),planner_discount=beta);  
  
//conduct stochastic simulations of the Ramsey problem  
stoch_simul(order=2,irf=20,periods=0,irf_plot_threshold=0)  
    LOG_pi LOG_h r LOG_C Z;  
evaluate_planner_objective;
```

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Discretionary policy equilibrium

- Planner re-optimizes each period and is expected to do so
→ no commitment!
- Denote with y_t the endogenous variables and with u_t the policy instruments
- Solution is only well-known for quadratic loss function

$$L_0 = \sum_{t=0}^{\infty} \beta^t (y_t' W y_t + u_t' Q u_t) \quad (32)$$

with linear constraints:

$$A_0 y_t = A_1 y_{t-1} + A_2 \mathbb{E}_t y_{t+1} + A_3 u_t + A_4 \mathbb{E}_t u_{t+1} + A_5 \varepsilon_t, \quad (33)$$

where ε_t are the structural shocks and W, Q, A_i are coefficient matrices

Solution structure

- We are looking for a Markov-perfect Stackelberg-Nash equilibrium:
 - Stackelberg leader: today's planner
 - Stackelberg followers: private sector agents and tomorrow's planner
- We conjecture our solution will be linear:

$$y_t = H_1 y_{t-1} + H_2 \varepsilon_t \quad (34)$$

$$u_t = F_1 y_{t-1} + F_2 \varepsilon_t, \quad (35)$$

where H_1 , H_2 , F_1 , F_2 are unknown matrices to be determined

- Plugging the guess into the constraints allows deriving a matrix equation whose fixed point corresponds to the solution

Dennis (2007) Solution algorithm

Step 1 For $i = 0$, initialize $H_1^{(i)}$, $H_2^{(i)}$, $F_1^{(i)}$, $F_2^{(i)}$

Step 2 Compute $D^{(i)} = A_0 - A_2 H_1^{(i)} - A_4 F_1^{(i)}$

Step 3 Solve the so-called Sylvester equation

$$P^{(i)} = W + \beta F_1^{(i)'} Q F_1^{(i)} + \beta H_1^{(i)} P^{(i)} H_1^{(i)} \quad (36)$$

Step 4 Set

$$\begin{aligned} F_1^{(i+1)} &= -(Q + A_3'(D^{(i)})^{-1} A_3)^{-1} A_3'(D^{(i)'})^{-1} P^{(i)} D^{(i)-1} A_1, \\ F_2^{(i+1)} &= -(Q + A_3'(D^{(i)'})^{-1} P^{(i)} D^{(i)-1} A_3)^{-1} A_3'(D^{(i)'})^{-1} P^{(i)} D^{(i)-1} A_5, \\ H_1^{(i+1)} &= D^{(i)-1} (A_1 + A_3 F_1^{(i+1)}), \\ H_2^{(i+1)} &= D^{(i)-1} (A_5 + A_3 F_2^{(i+1)}). \end{aligned}$$

Step 5 Go back to Step 2, until convergence.

In Dynare

- Quadratic loss function must be provided to Dynare in `planner_objective`
- Careful: no linear terms are allowed in the loss function and the variables in the loss function must be mean 0
- Instrument name and full steady state must be provided (not a conditional one)
- `discretionary_policy` to compute discretionary policy, generate IRFs, etc
→ similar to `ramsey_model` followed by `stoch_simul`

Dynare code

Ramsey_Example_discretionary.mod

- We need to use a quadratic loss function derived from the nonlinear planner objective:

```
steady_state_model;
  r=1/beta-1;
  Z=0;
  pi=(r+1)*beta;
  C=sqrt((1+1/theta*((1-beta)*(pi-1)*pi-(tau-1)/(theta-1))*(1-theta)))/
  (chi*(1+phi/2*(pi-1)^2));
  h=C*(1+phi/2*(pi-1)^2);
  y_nat=sqrt((theta-1)/theta*(1+tau)/chi);
  y_gap=log(C)-log(y_nat);
end;

planner_objective pi_hat^2*theta/((theta-1)/phi) + y_gap^2;

discretionary_policy(instruments=(r),planner_discount=beta);
```

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Program

- The general OSR problem involves the planner picking the optimal parameters γ of a policy rule

$$u_t = g(y_{t+1}, y_t, y_{t-1}, \varepsilon_t, \gamma) , \quad (37)$$

where u_t are the policy instruments to maximize the objective function

$$\max_{\gamma} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(y_t) \quad (38)$$

- Optimization is conducted subject to the private sector equilibrium conditions:

$$\mathbb{E}_t f(y_{t+1}, y_t, y_{t-1}, \varepsilon_t) = 0, \quad \forall t \geq 0 , \quad (39)$$

taking y_{-1} as given and imposing suitable transversality conditions

Program

- Up to Dynare 6, only special linear-quadratic (LQ) versions of the problem could be solved:

$$\min_{\gamma} \mathbb{E} [y_t' W y_t] \quad (40)$$

$$\text{s.t. } A_1 \mathbb{E}_t y_{t+1} + A_2 y_t + A_3 y_{t-1} + A_4 \varepsilon_t = 0, \quad (41)$$

where A_i are coefficient matrices obtained from using first-order perturbation

- W_t is the weighting matrix penalizing variances and covariances of the variables
→ specified in Dynare using the `optim_weights` block
- Loss function cannot be automatically derived from `planner_objective`
- Disadvantage: LQ setup restricted to first-order approximation, which is often insufficient for proper welfare evaluation (and comparisons to Ramsey)
- Advantage: LQ setup makes e.g. using analytical derivatives possible (`analytical_derivation-option`)

Legacy implementation in Dynare

- `osr_params` block specifies parameters γ to optimize over
- `optim_weights` block specifies weighting matrix W
- `osr_param_bounds` block sets bounds for parameter optimization
- `osr` command computes solution and results (analogously to `stoch_simul`)
- output and metadata located in fields `M_.osr` and `oo_.osr`

New implementation in Dynare 7

- Starting with Dynare 7 (and already available in the unstable version), a new implementation of the OSR algorithm is available
- Instead of `optim_weights` block a `planner_objective` command can be specified
- `osr` command computes solution at $\text{order} \geq 1$
- `evaluate_planner_objective` command computes welfare at the solution

Dynare code

Ramsey_Example_OS.R.mod

- Need to include the Taylor rule whose coefficient we optimize:

$$r - \bar{r} = \alpha(\pi - \pi^*) \quad (42)$$

```
model;
    [name='Euler equation']
    1/(1+r)=beta*C/(C(+1)*pi(+1));
    [name='Firm FOC']
    (tau-1/(theta-1))*(1-theta)+theta*(chi*h*C/(exp(Z))-1)
    =phi*(pi-1)*pi-beta*phi*(pi(+1)-1)*pi(+1);
    [name='Resource constraint']
    C*(1+phi/2*(pi-1)^2)=exp(Z)*h;
    [name='TFP process']
    Z=rho*Z(-1)+epsilon;
    [name='Taylor rule']
    r=pi_star/beta-1+alpha*(pi-pi_star);
end;
```

Dynare code

Ramsey_Example_OSR.mod

```
planner_objective log(C)-chi/2*h^2;

//define OSR parameters to be optimized
osr_params alpha;
//starting value for OSR parameter
alpha = 1.5;
//define bounds for OSR during optimization
osr_params_bounds;
    alpha, 0, 100;
end;
//compute OSR and provide output
osr(opt_algo=9,order=2,planner_discount=beta,irf_plot_threshold=0)
    LOG_pi LOG_h r LOG_C Z;
evaluate_planner_objective;
```

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